

Half-MS medium

Chemicals	Stock Conc.	Addition	Final Conc.	Location
MS basal salt (M407) [†]	—	1.22 g	0.5X	4°C (c)
Ammonium nitrate (NH ₄ NO ₃)	—	3.30 g	10.31 mM	VIII
Potassium nitrate (KNO ₃)	—	3.80 g	9.40 mM	toxic (D)
Potassium phosphate monobasic (KH ₂ PO ₄)	170 g / L	2 mL	0.63 mM	bench
MES hydrate	—	2 g	0.05% (w/v)	III
PPM [™]	100%	2 mL	0.05% (v/v)	4°C (a)
KOH ※	5 M	~ 730 µL	—	VII
ddH ₂ O	—	~ 4 L	—	—
Total volume		4 L		

[†] Murashige & Skoog Modified Basal Salt Mixture ([Product ID: M407](#)):

<https://phytottechlab.com/murashige-skoog-modified-basal-salt-mixture-m407.html>

Gelling agent → [Gelzan[™] \(G3251, CAS no.: 71010-52-1 \)](#) : 1.5 g / L (0.15%). Located between **I** and **II**.
Generally, we make 400 mL per bottle, so weigh (**600 ± 2**) **mg / 400 mL** for each bottle.

The final pH should be around **5.7 ~ 5.8**. Autoclave under **121°C** for **20 mins**.

※ KOH is used for pH adjustment. To prepare 5 M KOH solution, dissolve 330 g of 85% KOH in 900 mL ddH₂O, and then bring to 1 L.

Since KOH releases heat during dissolution, **gradually and slowly add the KOH granules into the water**.

Yoshida Medium

Yoshida's stock solution for root architecture media rice.

800X Yoshida macro-stock solution

800x macro stock	Grams / L	Location
NH_4NO_3	91.4	VIII
CaCl_2	88.6	IV
$\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ ※	40.3	VIII
K_2SO_4 §	71.4	VIII
$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ §	324	VIII

※ If use **$\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$** , then add **17.8** g / L;

§ K_2SO_4 and $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ will be easily contaminated, stored at 4°C;

800X Yoshida micro-stock solution

800x micro stock	Grams / L	Location
$\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$	1.5	VIII
$(\text{NH}_4)_6 \cdot \text{MO}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}$	0.074	VIII
H_3BO_3	0.934	VIII
$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	0.035	VIII
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	0.031	VIII
$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	7.7	VIII
Citric acid (monohydrate)	11.9	VIII

1X Yoshida medium (pH 5.8)

	1L	2L	4L	Location
800X Macro stock (mL)				
NH ₄ NO ₃	1.25	2.5	5	
NaH ₂ PO ₄ • H ₂ O	1.25	2.5	5	
K ₂ SO ₄	1.25	2.5	5	
CaCl ₂	1.25	2.5	5	
MgSO ₄ • 7H ₂ O	1.25	2.5	5	
800X Micro stock (mL)	1.25	2.5	5	
MES hydrate (g)	0.546	1.092	2.184	III

Hoagland medium

Modified from: <https://doi.org/10.1111/nph.20135> (Wang et al., 2024).

- The working solution is generally **0.5X**.
- To conduct P-deficiency experiment, reduce the K_2HPO_4 molarity to 4 μM .

500X Hoagland **macro**-stock solution

Chemical	M.W. (g / mol)	Molarity (mM)	1X solution (mg / L)	500X stock (g / L)	Storage
NH_4NO_3	80.04	5.6	448.22	224.11	VIII
K_2HPO_4	174.18	0.4	69.67	45.64	III
K_2SO_4	174.26	0.8	139.41	69.70	VIII
$\text{MgSO}_4\cdot 7\text{H}_2\text{O}$	246.47	0.8	197.18	98.59	VIII
CaCl_2	110.98	1.6	177.57	88.78	IV
KNO_3	101.10	0.8	80.88	40.44	Toxin (D)
Ferrous mixture	See below				

500X Ferrous Mixture

- The **$\text{Na}_2\text{EDTA}\cdot 2\text{H}_2\text{O}$** is slowly soluble in water at room temperature up to 0.26 M, which is approximately 96 mg in a final volume of 1 ml. The pH of this solution will be in the range of 4 to 6. EDTA salts are more soluble in water as the pH increases: the more EDTA there is in the salt form, the higher the pH of a water solution, and therefore, the higher the room temperature solubility. This can be achieved by a gradual addition of concentrated sodium hydroxide solution to the EDTA solution. A stock solution of 0.5 M at pH 8.5 is stable for months at 4°C. Solutions of EDTA may be autoclaved.
- The **$\text{FeSO}_4\cdot 7\text{H}_2\text{O}$** is sensitive to heat and light. It should be bluish green color. When it turn yellowish brown, that means it was already oxidized, do not use it.

Chemical	M.W. (g / mol)	Molarity (mM)	1X solution (mg / L)	500X stock (g / L)	Storage
$\text{Na}_2\text{EDTA}\cdot 2\text{H}_2\text{O}$	372.24	0.18	67.0	33.50	III
$\text{FeSO}_4\cdot 7\text{H}_2\text{O}$	278.01	0.18	50.04	25.02	IX

500X Hoagland **micro**-stock solution

Chemical	M.W. (g / mol)	Molarity (mM)	1X solution (mg / L)	500X stock (g / L)	Storage
H ₃ BO ₃	61.83	0.0230	1.4221	0.7110	VIII
MnCl ₂ •4H ₂ O	197.91	0.0045	0.8906	0.4453	VIII
CuSO ₄ •5H ₂ O	249.69	0.0003	0.0749	0.0375	VIII
ZnCl ₂	136.30	0.0015	0.2045	0.1022	N/A
Na ₂ MoO ₄ •2H ₂ O	241.95	0.0001	0.0242	0.0121	IX

The ZnCl₂ may be substituted with ZnSO₄•7H₂O.

	M.W. (g / mol)	Molarity (mM)	1X solution (mg / L)	500X stock (g / L)	Storage
ZnSO ₄ •7H ₂ O	287.56	0.0015	0.4313	0.2157	VIII